



RWANDA

Clean Cooking Financial Planning Tool

IN PARTNERSHIP WITH:



Global Energy Alliance
for People and Planet



July 2025



Presentation Overview

I. Project Description

II. Project Results

III. Tool Walkthrough

Project Description

Objectives, approach, and context of the tool development



Overview & Methodology

- **Objective:** Develop a tool to support clean cooking financial planning through scenario-based analysis based on the ICCPT for Rwanda.
- **Technologies:** Built using FastAPI backend served with Uvicorn, and Streamlit for interactive visual interface.
- **Context & Relevance:** Addresses the need for scalable, data-driven planning tools aligned with national energy strategies.
- **Initial Deployment:** Rwanda used as the baseline case for testing and development.
- **User-Centered Design:** Allows input customization and provides outputs for government planning, NGOs, or financial analysts.



User



Tool Interface



Inputs



Backend Engine



Outputs

Technical Architecture & Development Approach



1. Backend: FastAPI + Uvicorn

- **FastAPI:** High-performance web framework for building APIs with Python.
- **Uvicorn:** ASGI server for fast asynchronous request handling.
- **Role:** Manage business logic, calculations, data processing, and scenario modeling.

2. Frontend: Streamlit

- **Streamlit:** Interactive Python-based UI framework for data apps.
- **Role:** Provide editable tables, dynamic charts, and real-time visual feedback.

3. Modular & Scalable Design

- **Multi-Country Scenario Support:** Allows to define different national scenarios.
- **Scalable Architecture:** Designed to add or remove clean cooking technologies.
- **Flexible Integration:** Ready to incorporate new financing models, carbon methodologies, and energy policies without structural changes.

4. Development Workflow (Agile/Scrum)

- **Iterative sprints:** Regular releases with incremental features.
- **Modular tasks:** Each feature developed independently.
- **Version control:** GitHub with structured branching and merging.

5. Data Flow & Integration

User inputs → Streamlit UI → FastAPI backend → Calculations → Database/File storage → Visual output

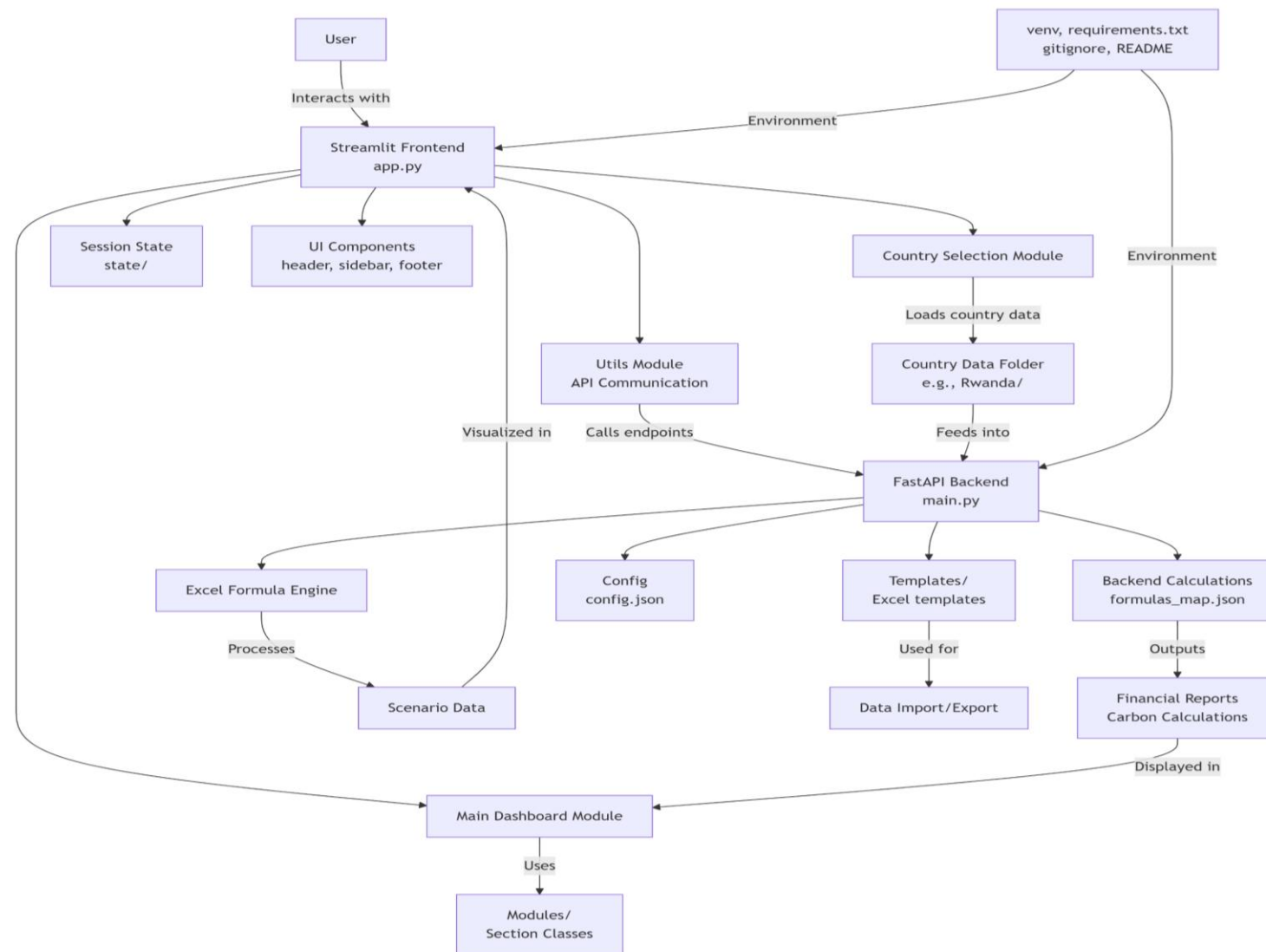
 **User**

  **Streamlit UI**
Country Selection
Dashboard
Input Forms

  **FastAPI Backend**
Calculations
Excel Engine
Scenario Management

  **Data & Outputs**
Financial Statements
Summary Dashboard
Carbon Credits

Module Interaction & System Architecture



Project Results

Tool functionalities, outputs delivered, and early outcomes



Key Achievements

- **Functional Tool:** End-to-end web tool with scenario input, backend calculations, and output visualization.
- **Editable Inputs:** Users define fuel types, time horizons, and financial assumptions via editable tables.
- **Automated Calculations:** Capital structure, financial statements (cash flow, P&L, and balance sheet) and potential carbon finance computed in backend.
- **Visual Dashboard:** Dynamic charts help interpret financial outcomes and carbon impact across scenarios.
- **Flexible & Scalable:** Architecture ready to integrate more countries, additional financing models, and new technologies for clean cooking adoption.

Electricity Financial Inputs

Inputs	Units	2023	2024	2025	2026	2027	2028
Expected non-technical losses	%	5	5	5	5	5	5
Trade receivables - Days of revenues	days	30	30	30	30	30	30
Trade payables - Days of OPEX costs	days	30	30	30	30	30	30

Manage Techno-Economic Inputs

Reset year range

Current year range: 2023 - 2034

CleanStep

Aligned

↓

↻

⬆

✖

↓

↻

⬆

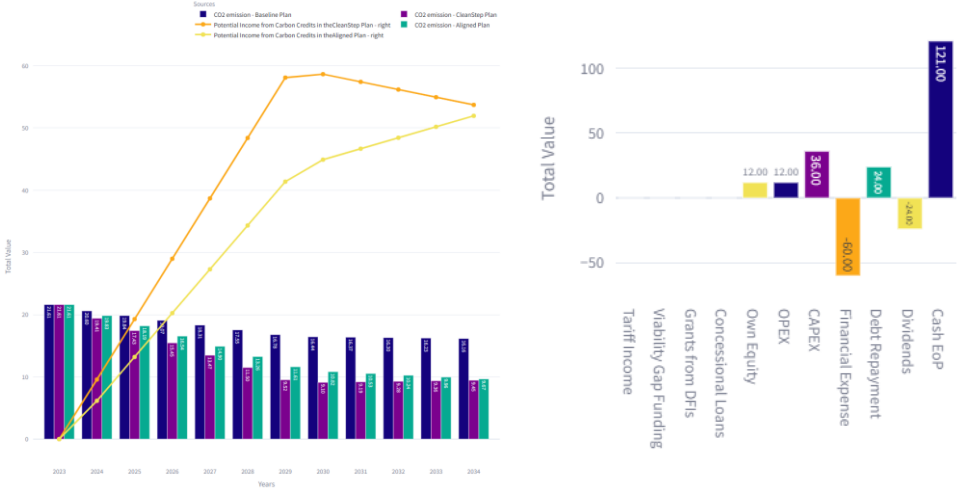
✖

+

 Create New Techno-Economic Model

Model Name

Create



Tool Walkthrough

Interaction flow from user inputs to final outputs



Welcome Screen

First page of the tool.

Appears immediately when the application is launched.

From here, the user moves forward by clicking the 'Select Scenario' button, which takes them to the page where they can choose a country as the scenario for the financial planning process.

The visual design—including images, layout, and logos—is still subject to changes based on design or specific user preferences.

Welcome to the Financial Clean Cooking Tool



Select Scenario



Country Selection

Second page of the tool.

User selects the country to be used as the scenario for the financial planning process.

The goal is to design a fully scalable solution that includes as many countries as desired, enabling users to adapt the tool to different national contexts and use cases over time.

Currently, only Rwanda is available, as it is being used as the baseline case for development and testing of the tool.

To add a new country, key national parameters such as the tax and inflation rates must be defined.

Users can download the scenario files for the selected country using the provided download button or delete the scenario entirely using the remove button.

Select a country scenario to begin modeling:

Rwanda

+ Add a new country

Select a country scenario to begin modeling:

Rwanda



Selected country: Rwanda



Start modeling

+ Add a new country

+ Add a new country

Country name

Tax Rate (%)

0

- +

Inflation (%)

0

- +

Add

Main Dashboard

Country selection

Financial Inputs

Manage Techno-Economic Models

Main dashboard of the tool.

User manages the techno-economic scenarios to be used for the financial planning.

The page presents the overall layout, featuring a sidebar on the left for navigation between sections, and the main content displayed at the center.

Manage Techno-Economic Inputs



Reset year range

Current year range: 2023 - 2034



CleanStep



Aligned



+ Create New Techno-Economic Model

Model Name

Create

Two example scenarios are already loaded into the tool. However, at this stage, the user can skip this step and go directly to the Financial Inputs section.

Financial Inputs – Add & Delete Fuel Types

Financial Inputs

Electricity

LPG

+ Add new fuel market

Delete fuel market

Electricity

Delete fuel

Add new fuel market

Enter name of the new technology:

Create market

Add new fuel market

Enter name of the new technology:

Ethanol

Create market

Delete fuel market

Ethanol

Are you sure you want to delete 'Ethanol'?

Yes, delete

Cancel

Financial Inputs

Electricity

Ethanol

LPG

+ Add new fuel market

Delete fuel market

Ethanol

Electricity

Ethanol

LPG

Carbon Crea

Delete fuel market

Ethanol

Delete fuel

Financial Inputs – Custom Input Tables

Country selection

Financial Inputs

Electricity Financial Inputs

LPG Financial Inputs

+ Add new fuel market

Delete fuel market

Electricity

Carbon Credits Financial Inputs

LPG Financial Inputs

Inputs	Units	2023	2024	2025	2026	2027	2028
Expected non-technical losses	%	5	5	5	5	5	5
Trade receivables - Days of revenues	days	30	30	30	30	30	30
Trade payables - Days of OPEX costs	days	30	30	30	30	30	30

Save

Reset

This section contains generic financial inputs for each fuel type, which are treated as platform-level parameters rather than model-specific values.

These inputs are consistent across all techno-economic scenarios within the same country context, reflecting standardized assumptions independent of individual modeling choices.

Use the Save button to store any changes or Reset to restore default values.

Manage Techno-Economic Scenarios – Define Scenario Timeframe

Set year range to apply uniformly to techno-economic models within the scenario, to ensure synchronized and consistent financial planning.

The screenshot displays the 'Manage Techno-Economic Inputs' interface. On the left is a sidebar with navigation options: 'Country selection', 'Financial Inputs', 'Manage Techno-Economic Models', and 'Outputs' (which includes 'Carbon Credits' and 'Summary Financing'). The main panel is titled 'Manage Techno-Economic Inputs' and shows the 'Current year range: 2023 - 2034'. Below this, there are two rows of model cards: 'CleanStep' and 'Aligned'. Each card has four action buttons: a download icon, an upload icon, a refresh icon, and a delete icon (a red 'X'). A red box highlights the 'Reset year range' button, which is located above the model cards. A dialog box is open, showing the same 'Reset year range' button and the current year range. Below the button, a yellow warning message states: 'The year range defined previously will be removed, and all techno-economic models associated with this scenario will be deleted.' At the bottom of the dialog are two buttons: 'Confirm reset' (with a green checkmark) and 'Cancel' (with a red 'X'). A mouse cursor is pointing at the 'Cancel' button.

Manage Techno-Economic Inputs

Current year range: 2023 - 2034

Reset year range

CleanStep

Aligned

+ Create New Techno-Economic Model

Model Name

Create

Current year range: 2023 - 2034

Reset year range

The year range defined previously will be removed, and all techno-economic models associated with this scenario will be deleted.

☒ Confirm reset

☐ Cancel

Manage Techno-Economic Scenarios – Add & Delete

Reset year range

Current year range: 2023 - 2034

CleanStep

↓

✎

📁

✖

Aligned

↓

✎

📁

✖

+ Create New Techno-Economic Model

Model Name

New Model

Create

Model 'New Model' created successfully for country 'Rwanda'.

CleanStep

↓

✎

📁

✖

Aligned

↓

✎

📁

✖

New Model

↓

✎

📁

✖

⚠ Are you sure you want to delete New Model? This action cannot be undone.

✔ Confirm delete

✖ Cancel

CleanStep

↓

✎

📁

✖

Aligned

↓

✎

📁

✖

The user creates a new scenario by entering a unique name. The year range is inherited from the previously defined timeframe, ensuring consistency across all models for coherent financial planning.

Scenarios can be deleted using the remove button (✖), deleting all associated files and data related to that model.

Manage Techno-Economic Scenarios – Import & Export

Manage Techno-Economic Inputs

Reset year range Current year range: 2023 - 2034

CleanStep

Aligned

Download files

Rwanda_Aligned_files.zip
[Abrir archivo](#)

Manage Techno-Economic Inputs

Reset year range Current year range: 2023 - 2034

CleanStep

Aligned

Download template

upload-Aligned.xlsx
[Abrir archivo](#)

Manage Techno-Economic Inputs

Reset year range Current year range: 2023 - 2034

CleanStep

Aligned

Upload template

Upload file for Aligned

Drag and drop file here
Limit 200MB per file • XLSX

Browse files

Download Options:

- **ZIP Archive:** Download all files related to the selected scenario.
- **Excel Template:** Export a standardized Excel template to prepare data offline.

Upload Option:

Upload the same completed Excel template to automatically populate data tables, ensuring correct formatting and eliminating manual entry.

Techno-Economic Scenarios

- Inputs

Manage Techno-Economic Inputs

Reset year range Current year range: 2023 - 2034

CleanStep

Aligned

This section displays **two layers of data**:

- **Financial Inputs (Non-Editable):**
Predefined inputs for each fuel type, set in the *Financial Inputs* section. These values are fixed per fuel and serve as the baseline for all scenarios.
- **Techno-Economic Inputs (Editable):**
Model-specific inputs for each technology within the selected scenario.
Tables are fully editable, allowing users to modify values as needed.

Country selection

Financial Inputs

Manage Techno-Economic Models

Techno-Economic Model: Aligned

Inputs

Techno-Economic Inputs

Electricity & E-Cooking

Electricity (Low access)

LPG

Tariffs

Upstream Costs of Energy

CAPEX Fuel Market

Design Capital Structure

Financial Plan

Electricity & E-Cooking Inputs

Inputs	Units	2023	2024	2025	2026	2027
Expected non-technical losses	%	5	5	5	5	5
Trade receivables - Days of revenues	days	30	30	30	30	30
Trade payables - Days of OPEX costs	days	30	30	30	30	30

System	Inputs	Units	2023	2024	2025	2026	2027
GRID	Demand	kWh	1473.57	1578.81	1718.68	1846.2	1962.0
GRID	CAPEX - Growth	M\$	242.2	294.74	291.79	288.87	285.9
GRID	CAPEX - Maintenance	M\$	0	0	0	0	0
GRID	OPEX	M\$	48.2	53.31	59.62	65.8	71.8
GRID	D&A	years	35				
OFF-GRID	Demand	kWh	368.39	370.34	377.27	378.14	373.7
OFF-GRID	CAPEX - Growth	M\$	39.21	49.67	49.17	48.68	48.1
OFF-GRID	CAPEX - Maintenance	M\$	0	0	0	0	0
OFF-GRID	OPEX	M\$	56.13	63.13	71.77	80.24	88.5
OFF-GRID	D&A	years	25				

Save

Reset

Save button stores changes and Reset button clears all editable fields and restore the table to default state of zeros.

Techno-Economic Scenarios - Tariffs

Tariff	2023	2024	2025	2026	2027	2028	2029	2030	2031
M\$/kWh	0	0	0	0	0	0	0	0	0

Users can define tariffs in two ways:

- **Automatic Calculation:**
Set an *initial tariff value* and automatically project future tariffs based on the country's predefined inflation rate.
- **Manual Entry:**
Enter *year-by-year tariff values* manually for full control over the pricing scenario.

Tariff	2023	2024	2025	2026	2027	2028	2029	2030	2031
M\$/kWh	0.15	0.16	0.17	0.17	0.18	0.19	0.2	0.21	

Changes are saved using the Save button or reset to restore default values.

Techno-Economic Scenarios - Upstream Costs of Energy

Financial Inputs

Manage Techno-Economic Models

Techno-Economic Model: Aligned

Inputs

Techno-Economic Inputs

Tariffs

Upstream Costs of Energy

Electricity

LPG

CAPEX Fuel Market

No upstream configuration found for Electricity. Please set it up.

Upstream Configuration for Electricity

Country: Rwanda | Model: Aligned

Year Range: 2023 - 2034

Inflation Rate: 5%

How would you like to define the upstream values?

☐ Initial value + inflation (autofill based on inflation rate)

☒ Annual values (enter each year separately)

Electricity Upstream Editor

Upstream	2023	2024	2025	2026	2027	2028	2029	2030
MS/kWh	0	0	0	0	0	0	0	0

Save

Upstream costs are defined following the same method as tariffs:

- Set an **initial value** for automatic inflation-based projection, or
- Enter **annual values manually** for full control.

Financial Inputs

Manage Techno-Economic Models

Techno-Economic Model: Aligned

Inputs

Techno-Economic Inputs

Tariffs

Upstream Costs of Energy

Electricity

LPG

CAPEX Fuel Market

Design Capital Structure

Financial Plan

Outputs

Financial Statements

No upstream configuration found for Electricity. Please set it up.

Upstream Configuration for Electricity

Country: Rwanda | Model: Aligned

Year Range: 2023 - 2034

Inflation Rate: 5%

How would you like to define the upstream values?

☒ Initial value + inflation (autofill based on inflation rate)

☐ Annual values (enter each year separately)

Initial Upstream Value

Enter initial upstream value:

0,08

Apply Inflation Calculation

Electricity Upstream Editor

Upstream	2023	2024	2025	2026	2027	2028	2029	2030
MS/kWh	0.08	0.08	0.09	0.09	0.1	0.1	0.11	0.11

Save

Reset

Save or reset entries as needed.

CAPEX Fuel Markets

This section displays CAPEX details for each fuel type defined in the *Financial Inputs* section. The tables are not editable, as they rely on user-provided data from the *Techno-Economic Inputs* section and are calculated automatically by the backend.

Techno-Economic Model: Aligned

Inputs

Techno-Economic Inputs

Tariffs

Upstream Costs of Energy

CAPEX Fuel Market

Electricity & E-Cooking

Electricity (Low access)

LPG

LPG - CAPEX

Type	Units	2023	2024	2025	2026	2027	2028	2029
Growth CAPEX	M\$	3.71	6.82	6.79	6.76	6.73	6.7	5.16
Acc. Growth CAPEX	M\$	3.71	10.53	17.32	24.08	30.81	37.51	42.68
Maintenance CAPEX	M\$	0	0	0	0	0	0	0
Acc. Maintenance CAPEX	M\$	0	0	0	0	0	0	0
Total CAPEX	M\$	3.71	6.82	6.79	6.76	6.73	6.7	5.16
Acc. Total CAPEX	M\$	3.71	10.53	17.32	24.08	30.81	37.51	42.68
Total Depreciation	years	15						
Total Annual Depreciation	M\$	0.25	0.7	1.15	1.61	2.05	2.5	2.85
RAB	M\$	3.46	9.58	15.22	20.37	25.05	29.25	31.57

Design Capital Structure

This section integrates user-defined inputs—such as equity and grant percentages—with backend-calculated financing requirements, presenting a complete funding plan for the selected scenario.

Calculated Financial Tables

Financiation	Units	Amount
How much to be financed	M\$	0

Total	Units	Amount
Equity	M\$	0
Grants	M\$	0
Debt	M\$	0
WACC	%	0

Techno-Economic Model: Aligned

Inputs

Techno-Economic Inputs

Tariffs

Upstream Costs of Energy

CAPEX Fuel Market

Design Capital Structure

Financial Plan

Electricity & E-Cooking

Electricity (Low access)

LPG

Outputs

Financial Statements

Carbon Credits

Capital Structure for LPG - Input Tables

Division	Units	Amount
1. Equity	%	40
Cost of Equity	%	12
2. Grants	%	0
3. Debt		
Cost of Debt	%	8
Grace period	years	6
Amortization period	years	25

FFSS - Outputs

Type

Units

2023

Calculated debt requirement

Annual debt needs

ok

Calculated debt variation

How much to increase

M\$

0

FFSS - Inputs

Type

Units

2023

2024

User-defined debt increase

Debt increase

M\$

0

0

Save

Reset

Calculate Financial Tables

Financial Statements

This section constitutes one of the tools’s key outputs. It provides the user with financial statements and key indicators, including the Profit and Loss statement, Balance Sheet, Cash Flow statement, PP&E CAPEX, Working Capital calculations, Equity Schedule, and Capital Structure.

Techno-Economic Model: Aligned

Inputs

Techno-Economic Inputs



Tariffs



Upstream Costs of Energy



CAPEX Fuel Market



Design Capital Structure

Financial Plan



Outputs

Financial Statements



Electricity & E-Cooking

Electricity (Low access)

Electricity (Only E-Cooking)

LPG

Electricity & E-Cooking - FFSS

Profit & Loss	Type	Units	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Total Revenues		M\$	276.29	319.38	371.76	400.32	450.34	499.62	547.89	607.92	664.88	720.08	773.33	857.43
Total Revenues	% Growth	%	0	15.59	16.4	7.68	12.49	10.94	9.66	10.96	9.37	8.3	7.39	10.88
Tariff Income - Energy & SHS		M\$	276.29	319.38	371.76	400.32	450.34	499.62	547.89	607.92	664.88	720.08	773.33	857.43
Tariff Income - Energy & SHS	% Growth	%	0	15.59	16.4	7.68	12.49	10.94	9.66	10.96	9.37	8.3	7.39	10.88
Grants		M\$	0	0	0	0	0	0	0	0	0	0	0	0
Grants	% Growth	%	0	0	0	0	0	0	0	0	0	0	0	0
Grants	% Revenues	%	0	0	0	0	0	0	0	0	0	0	0	0
Long term subsidies		M\$	0	0	0	0	0	0	0	0	0	0	0	0
Long term subsidies	% Growth	%	0	0	0	0	0	0	0	0	0	0	0	0
Long term subsidies	% Revenues	%	0	0	0	0	0	0	0	0	0	0	0	0
Annual Cost of Service		M\$	22216.79	458.92	28597.12	597.23	675.22	752.01	828.12	909.64	982.78	1053.93	1122.9	1219.02
Annual Cost of Service	% Growth	%	0	-97.93	6131.45	-97.91	13.06	11.37	10.12	9.84	8.04	7.24	6.54	8.56
Annual Cost of Service	% Revenues	%	8041	143.69	7692.37	149.19	149.94	150.52	151.15	149.63	147.81	146.36	145.2	142.17
Upstream Cost of Energy		M\$	22103.52	319.38	28428.66	400.32	450.34	499.62	547.89	607.92	664.88	720.08	773.33	857.43
Upstream Cost of Energy	% Growth	%	0	-98.56	8801.27	-98.59	12.49	10.94	9.66	10.96	9.37	8.3	7.39	10.88

Balance Sheet	Units	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Non-current assets	M\$	287.86	642.59	982.39	1307.42	1617.82	1913.74	2235.34	2581.38	2663.03	2983.68	3288.23	3452.52
PP&E	M\$	287.86	642.59	982.39	1307.42	1617.82	1913.74	2235.34	2581.38	2663.03	2983.68	3288.23	3452.52
Current assets	M\$	1830.86	42.69	2355.89	2351.1	4655.91	6960.63	9265.21	11572.3	13879.32	1607...	1821...	2037...
Cash and equivalents	M\$	0	0	0	2297.99	4595.98	6893.96	9191.95	11489.94	13787.93	15976.2	1810...	2025...
Trade receivables	M\$	22.71	26.25	30.56	32.9	37.01	41.06	45.03	49.97	54.65	59.18	63.56	70.47
Inventories	M\$	1808.15	16.44	2325.34	20.21	22.92	25.6	28.23	32.4	36.74	40.95	45.01	51.61
Total Assets	M\$	2118.72	685.28	3338.28	3658.52	6273.73	8874.37	11500.55	14153.68	16542.35	1906...	2150...	23830

Carbon Credits

This section is another key output of the platform, divided into two parts:

Emissions Input: where users enter CO₂ emissions for the Baseline and each Techno-Economic Scenario.

Comparative Results: automatically generated by the backend after clicking “Calculate Carbon Credits”, the tool computes and displays the avoided emissions and potential carbon credit income.

Techno-Economic Model: CleanStep

Inputs

Techno-Economic Inputs

Tariffs

Upstream Costs of Energy

CAPEX Fuel Market

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Financial Plan

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Financial Statements

Carbon Credits

Carbon Credits - Input Emissions

CO2 Inputs	Units	2023	2024	2025	2026	2027	2028
CO2 emitted - Baseline scenario	CO2Eq MT/y	21.61	20.6	19.84	19.07	18.31	17.5
CO2 emitted - Aligned scenario	CO2Eq MT/y	21.61	19.83	18.19	16.54	14.9	13.2
CO2 emitted - CleanStep scenario	CO2Eq MT/y	21.61	19.41	17.43	15.45	13.47	11.1

Save

Reset

Calculate Carbon Credits

Carbon Credits

	Units	2023	2024	2025
Δ▲Aligned vs. Baseline	CO2Eq MT/y	0	0.77	1.65
Δ▲CleanStep vs. Baseline	CO2Eq MT/y	0	1.2	2.41
CO2 equivalent avoided in the Aligned	CO2Eq MT/y	0	0.61	1.32
CO2 equivalent avoided in the CleanStep	CO2Eq MT/y	0	0.96	1.93
Potential income from Carbon Credits in the Aligned Plan - right	M USD	0	6.15	13.19
Potential income from Carbon Credits in the CleanStep Plan - right	M USD	0	9.57	19.28

Summary Financing

This section provides an **intuitive visual overview** of financial and carbon outcomes, with dynamic charts generated from both user inputs and backend-calculated results.

Techno-Economic Model: CleanStep

Inputs

Techno-Economic Inputs

Tariffs

Upstream Costs of Energy

CAPEX Fuel Market

Design Capital Structure

Financial Plan

Outputs

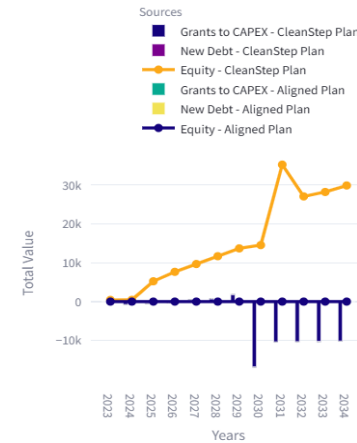
Financial Statements

Carbon Credits

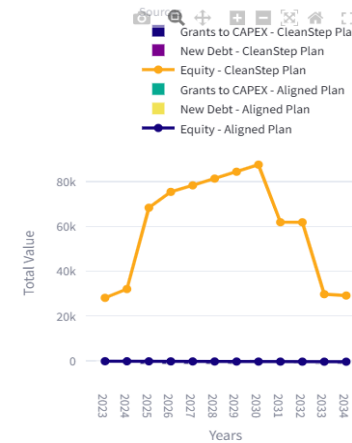
Summary Financing

Summary Financing Dashboard

Accumulated Capital Structure for the Electricity (Only E-Cooking) Sector



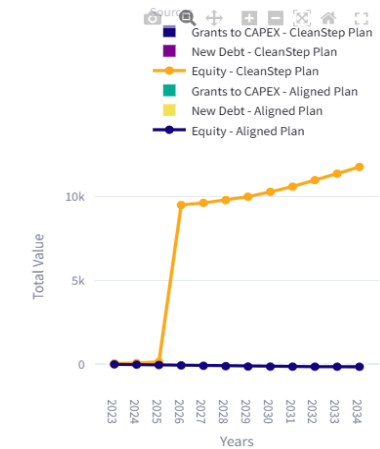
Accumulated Capital Structure for the Electricity & E-Cooking Sector



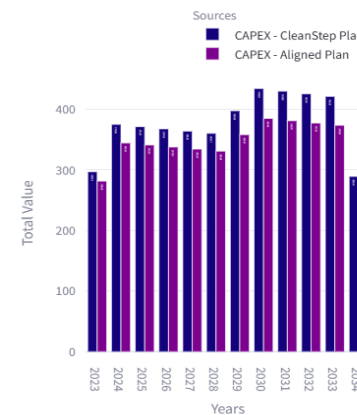
Accumulated Capital Structure for the Electricity (Low access) Sector



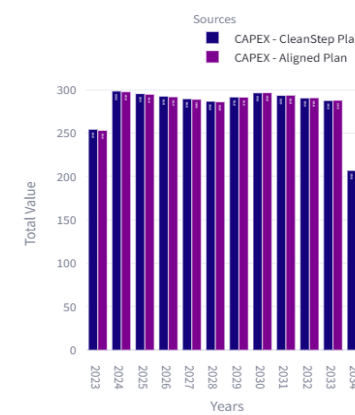
Accumulated Capital Structure for the LPG Sector



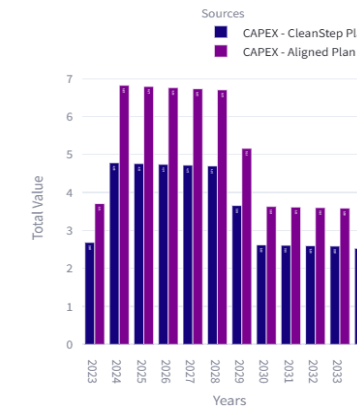
CAPEX Investments for the Electricity & E-Cooking Sector



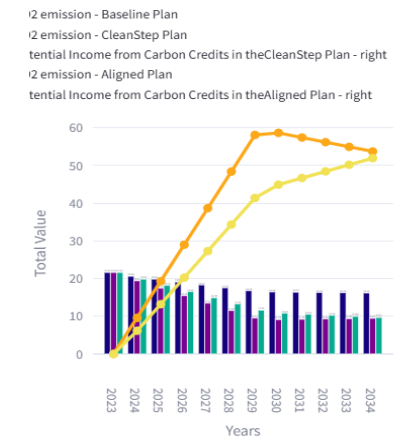
CAPEX Investments for the Electricity (Low access) Sector



CAPEX Investments for the LPG Sector



CO2 emission [CO2Eq MT/y]



Thank you

